

Maxwell Theory from a Hermitian Complex Line Bundle

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1 Spacetime

We begin with a smooth, oriented, four-dimensional manifold

$$M.$$

To formulate classical electromagnetism, we equip M with a Lorentzian metric

$$g \in \Gamma(S^2 T^* M)$$

of signature

$$(-, +, +, +).$$

Thus, at each point $x \in M$, the metric defines a nondegenerate symmetric bilinear form

$$g_x : T_x M \times T_x M \longrightarrow \mathbb{R}.$$

We assume that M is oriented and time-oriented.

The metric and orientation determine the Hodge star operator

$$* : \Omega^k(M) \longrightarrow \Omega^{4-k}(M).$$

For differential forms $\alpha, \beta \in \Omega^k(M)$, the Hodge star is characterized by

$$\alpha \wedge * \beta = \langle \alpha, \beta \rangle_g \text{ vol}_g,$$

where vol_g denotes the metric volume form.

The Hodge star is the part of Maxwell theory that depends explicitly on the geometry of spacetime.

2 Complex Line Bundles

Definition 2.1 (Complex line bundle). *A complex line bundle over M is a smooth vector bundle*

$$\pi : L \longrightarrow M$$

whose fibers

$$L_x = \pi^{-1}(x)$$

are one-dimensional complex vector spaces.

Equivalently,

$$\dim_{\mathbb{C}} L_x = 1$$

for every $x \in M$.

Locally, every complex line bundle is trivial. Thus for every $x \in M$, there exists an open neighborhood $U \subset M$ such that

$$L|_U \cong U \times \mathbb{C}.$$

A smooth section of L is a smooth map

$$\psi : M \longrightarrow L$$

such that

$$\pi \circ \psi = \text{id}_M.$$

The space of smooth sections is denoted by

$$\Gamma(L).$$

3 Hermitian Structure

Definition 3.1 (Hermitian structure). *A Hermitian structure on L is a smoothly varying family of positive-definite Hermitian inner products*

$$h_x : L_x \times L_x \longrightarrow \mathbb{C}.$$

Thus

$$h_x(v, w) = \overline{h_x(w, v)}$$

and

$$h_x(v, v) > 0$$

for every nonzero $v \in L_x$.

A complex line bundle equipped with such a metric is called a *Hermitian complex line bundle*.

The Hermitian structure reduces the structure group of the line bundle from

$$\mathbb{C}^\times$$

to

$$U(1).$$

Consequently, electromagnetism may be interpreted geometrically as a $U(1)$ gauge theory.

4 Connections

Values of a section at distinct spacetime points belong to different vector spaces:

$$\psi(x) \in L_x, \quad \psi(y) \in L_y.$$

There is no canonical way to compare these vectors. A connection provides an infinitesimal rule for doing so.

Definition 4.1 (Connection). *A connection on L is a $\mathbb{C}$ -linear map*

$$\nabla : \Gamma(L) \longrightarrow \Omega^1(M; L)$$

satisfying the Leibniz rule

$$\nabla(f\psi) = df \otimes \psi + f\nabla\psi$$

for every

$$f \in C^\infty(M, \mathbb{C}), \quad \psi \in \Gamma(L).$$

Definition 4.2 (Hermitian connection). *A connection ∇ is compatible with the Hermitian structure if*

$$d h(\psi, \phi) = h(\nabla\psi, \phi) + h(\psi, \nabla\phi)$$

for all sections $\psi, \phi \in \Gamma(L)$.

Choose a local unitary frame e over an open subset $U \subset M$. Every section can then be written locally as

$$\psi = \psi_U e.$$

A Hermitian connection takes the local form

$$\nabla = d + iA,$$

where

$$A \in \Omega^1(U; \mathbb{R}).$$

The real one-form A is the electromagnetic gauge potential.

5 Gauge Transformations

A different unitary frame is related to the original one by

$$e' = e^{-i\lambda}e,$$

where

$$\lambda : U \longrightarrow \mathbb{R}.$$

Under this change of frame, the local gauge potential transforms as

$$A \longmapsto A + d\lambda.$$

This is precisely the usual $U(1)$ gauge transformation.

Therefore, the potential A is not itself an invariant geometric object. Rather, it is a local representative of a connection on the line bundle.

6 Curvature

Definition 6.1 (Curvature). *The curvature of a connection ∇ is the two-form with values in $\text{End}(L)$ defined by*

$$F_\nabla = \nabla^2.$$

For a Hermitian line bundle, the curvature takes values in the Lie algebra

$$\mathfrak{u}(1) = i\mathbb{R}.$$

With the convention

$$\nabla = d + iA,$$

we obtain

$$F_\nabla = i dA.$$

It is convenient to define the real electromagnetic field-strength two-form by

$$F = dA.$$

Thus

$$F_\nabla = iF.$$

Under a gauge transformation,

$$A \mapsto A + d\lambda,$$

we have

$$F \mapsto d(A + d\lambda) = dA = F.$$

Hence F is gauge invariant.

7 The First Maxwell Equation as a Bianchi Identity

Since locally

$$F = dA,$$

we immediately obtain

$$dF = d^2A = 0.$$

More intrinsically, this is the Abelian form of the Bianchi identity.

Thus the first half of Maxwell's equations is not a dynamical equation. It follows from the geometric fact that the electromagnetic field is the curvature of a $U(1)$ connection:

$$\boxed{dF = 0.}$$

8 The Maxwell Action

The dynamics of the connection are determined by an action functional.

For the free electromagnetic field, define

$$S[A] = -\frac{1}{2} \int_M F \wedge *F,$$

where

$$F = dA.$$

Equivalently, in local coordinates,

$$S[A] = -\frac{1}{4} \int_M F_{\mu\nu} F^{\mu\nu} \sqrt{|\det g|} d^4x.$$

Here

$$F = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu,$$

with

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

9 Variation of the Maxwell Action

Consider a variation

$$A \mapsto A + \varepsilon a,$$

where

$$a \in \Omega^1(M)$$

has compact support.

Then

$$F \mapsto F + \varepsilon da,$$

so

$$\delta F = da.$$

The variation of the action is

$$\begin{aligned} \delta S &= -\frac{1}{2} \delta \int_M F \wedge *F \\ &= - \int_M da \wedge *F. \end{aligned}$$

Using

$$d(a \wedge *F) = da \wedge *F - a \wedge d *F,$$

we obtain

$$da \wedge *F = d(a \wedge *F) + a \wedge d *F.$$

Therefore,

$$\delta S = - \int_M d(a \wedge *F) - \int_M a \wedge d *F.$$

Assuming either that M has no boundary or that the variation vanishes sufficiently rapidly at the boundary, the first term vanishes by Stokes' theorem. Hence

$$\delta S = - \int_M a \wedge d *F.$$

Since a is arbitrary, the Euler–Lagrange equation is

$$\boxed{d * F = 0.}$$

10 Vacuum Maxwell Equations

Combining the geometric identity

$$dF = 0$$

with the Euler–Lagrange equation

$$d * F = 0,$$

we obtain the vacuum Maxwell equations:

$$\begin{aligned} dF &= 0, \\ d * F &= 0. \end{aligned}$$

The two equations have conceptually different origins:

$$\begin{array}{ccc} \text{connection} & \longrightarrow & \text{curvature} \\ \nabla & \longrightarrow & F \\ & & \downarrow \\ & & dF = 0, \end{array}$$

whereas

$$\begin{array}{ccc} g & \longrightarrow & * \\ & & \downarrow \\ F & \longrightarrow & S[F] \\ & & \downarrow \\ & & d * F = 0. \end{array}$$

Thus

$$dF = 0$$

is a geometric identity, while

$$d * F = 0$$

is a dynamical field equation.

11 Electric and Magnetic Fields

In Minkowski spacetime with coordinates

$$(x^0, x^1, x^2, x^3) = (t, x, y, z),$$

the electromagnetic two-form can be decomposed as

$$F = E_i dx^i \wedge dt + \frac{1}{2} \epsilon_{ijk} B^k dx^i \wedge dx^j.$$

Then

$$dF = 0$$

is equivalent to

$$\nabla \cdot \mathbf{B} = 0$$

and

$$\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0.$$

Similarly,

$$d * F = 0$$

is equivalent to

$$\nabla \cdot \mathbf{E} = 0$$

and

$$\nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} = 0$$

in units where

$$c = \epsilon_0 = \mu_0 = 1.$$

12 Coupling to an Electric Current

Let

$$J \in \Omega^1(M)$$

denote the electric four-current one-form.

The interaction term may be written as

$$S_{\text{int}} = \int_M A \wedge *J.$$

The total action is therefore

$$S[A] = -\frac{1}{2} \int_M F \wedge *F + \int_M A \wedge *J.$$

Variation with respect to A gives

$$\boxed{d * F = *J.}$$

Together with the Bianchi identity,

$$\boxed{dF = 0,}$$

the Maxwell equations with sources are

$$\boxed{\begin{aligned} dF &= 0, \\ d * F &= *J. \end{aligned}}$$

Applying d to the second equation yields

$$d * J = 0.$$

This is the differential-form expression of electric charge conservation.

13 Topology of the Electromagnetic Field

If the line bundle L is globally trivial, one may choose a global potential

$$A \in \Omega^1(M)$$

and write globally

$$F = dA.$$

For a nontrivial line bundle, however, a global potential need not exist. Instead, one has local potentials

$$A_\alpha \in \Omega^1(U_\alpha)$$

related on overlaps by gauge transformations

$$A_\beta = A_\alpha + d\lambda_{\alpha\beta}.$$

Nevertheless, the curvature forms agree:

$$dA_\alpha = dA_\beta.$$

Consequently they define a global two-form

$$F \in \Omega^2(M).$$

With the appropriate normalization, its de Rham cohomology class satisfies

$$\left[\frac{F}{2\pi} \right] = c_1(L)_{\mathbb{R}},$$

where

$$c_1(L) \in H^2(M; \mathbb{Z})$$

is the first Chern class of the line bundle.

Thus the topology of the electromagnetic gauge bundle is encoded in the cohomology class of the electromagnetic field strength.

14 Geometric Summary

The construction may be summarized as

$$\boxed{(M, g) \longrightarrow (L, h) \longrightarrow \nabla \longrightarrow F_\nabla \longrightarrow S[\nabla].}$$

More explicitly,

$$\begin{aligned} M &= \text{spacetime,} \\ L \rightarrow M &= \text{Hermitian complex line bundle,} \\ \nabla &= U(1) \text{ connection,} \\ A &= \text{local connection one-form,} \\ F &= \text{curvature / electromagnetic field,} \\ S &= -\frac{1}{2} \int_M F \wedge *F. \end{aligned}$$

The Maxwell equations then arise from two different geometric principles:

$$\boxed{\underbrace{dF = 0}_{\text{Bianchi identity}}, \quad \underbrace{d * F = *J}_{\text{Euler-Lagrange equation}}.}$$

In this formulation, classical electromagnetism is therefore the gauge theory of connections on Hermitian complex line bundles.

15 Relation to Yang–Mills Theory

The preceding construction is the Abelian case of Yang–Mills theory.

For electromagnetism the structure group is

$$U(1),$$

which is Abelian, and locally

$$F = dA.$$

For a non-Abelian Lie group G , a connection is locally represented by a Lie-algebra-valued one-form

$$A \in \Omega^1(M; \mathfrak{g}),$$

and its curvature becomes

$$F_A = dA + A \wedge A.$$

The Yang–Mills action is

$$S_{\text{YM}}[A] = -\frac{1}{2} \int_M \text{tr} (F_A \wedge *F_A) .$$

Thus Maxwell theory is the simplest example of the general geometric principle

gauge field = connection, field strength = curvature.
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